Matched-Filter Recovery of Pasted Noise and a Twelve-Member Sound-Event Ensemble for Indic Speech Noise Event Detection

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Code, heads and submissions: https://github.com/syehsyoh-dokling/NoiseDetection

***Abstract*—Track 1 asks for the onset and offset of every noise event in Vaani field recordings, scored by an event-based F1 (both boundaries within 20 % of the reference duration, at least 50 ms) plus a per-clip segment Dice. Our final submission (Codabench submission 994586) placed 5th of 82 teams with a combined score of 1.600895 (F1 0.780264, Dice 0.820631; maximum 2.0). The synthetic clips appear to be clean speech with noise samples pasted from a finite library; the validation release ships the clean references, so mixture minus reference recovers the pasted waveforms, and a normalised cross-correlation (NCC) matched filter over 1,198 such templates locates them in the test clips. Natural clips are handled by a frame-level sound-event detector (SED): a weighted mean of twelve members, each a bidirectional GRU (BiGRU) head over a frozen or fine-tuned Whisper, AST, BEATs or Dasheng encoder, decoded with per-event peak-relative boundaries. We report the leaderboard lineage, out-of-fold component scores, and the reproduction of the submission after the original fine-tuned checkpoints were lost.**

I. Training Data and Preprocessing

Vaani data. The released validation set holds 2,501 clips (3.51 h, 16 kHz mono): 1,806 natural field recordings and 695 synthetic clips, each synthetic clip with its clean reference, from 28 states, 141 state/district pairs and 27 languages, with 5,157 labelled events in seven top-level categories (human non-speech 1,924; vehicle/traffic 1,281; animal 1,142; baby/child 496; phone/signal/alarm 188; singing/music 68; appliance/machine 58). The test set holds 5,517 unlabelled clips; no test labels were released or accessed.

External data (permitted by the rules [1], which allow additional public or proprietary data). We used the Gold tier of the Vaani noise-event dataset [2], [3] (10,772 clips, 19,414 verified-timestamp events; CC BY 4.0, gated access which we hold) for every member except wsA and wlP, and its Silver tier (unverified timestamps) for two members, wlHS (14,529 clips extracted from the Hugging Face release) and bFS (the first 12,000 clips of the Kaggle copy ayanokoji2028/vaani-noise-hf of [2], HF file names), at loss weight 0.4 (member tags are defined in Sec. II). Overlap with the competition data was checked by Vaani file stem, a unique clip id: Gold shares no clip with the test set (0 of 5,517) or the validation set (0 of 2,501) and no recording session (file-name prefix) with any test clip; 13 Gold clips share a recording session with a validation clip. Both Silver extractions exclude every validation and test file name by construction. Speaker/district overlap with the test set was not checked (domain, not label, leakage).

Test-side information. Two uses are disclosed. (a) The leaderboard score of each upload was the final selection criterion (Sec. III); Codabench [1] lists 47 uploads by our team (46 scored; one, 995069 on 17 Oct after the final, failed and was not scored) and our first upload (921278, 10 Sep) is no longer listed, so at least 48 were made, under a quota of 5 per day and 100 in total. (b) Member wlP was trained on pseudo-labels [4] that three validation-only Whisper heads (small, medium and large, weighted 1/1/2) produced on the 3,353 test clips without a library hit, a transductive use of unlabelled test audio; the organisers list semi-supervised use of unlabelled Vaani audio as a novelty direction, which we read as covering these clips. The fold-f student only sees labels from teachers that never saw fold f (leak-free OOF); Table II reports the ensemble without it.

Template library. Outside the pasted spans ($\pm$150 ms) the residual (mixture minus clean reference) of a synthetic validation clip is zero at most samples (per-clip median 0, overall maximum 4.4×10^{-3} at full scale 1), so every labelled synthetic event cut from the residual is the pasted noise sample. Near-duplicates (NCC >0.995) were merged, leaving 1,198 templates, each with its source clips for leave-one-clip-out (LOO) evaluation.

Features. Every detector works on a 20 ms frame grid. For each encoder, the tokens of three or four selected layers are averaged over frequency (AST, BEATs, Dasheng), layer-normalised per frame, averaged over layers and interpolated from their centres onto the grid; Whisper's 20 ms encoder frames are used directly, at the true clip length. Targets are eight binary channels per frame (any noise plus the seven categories): 24 % of natural reference events overlap in time with an event of another category, which the event-F1 term scores separately.

Validation protocol. All model selection used 5-fold cross-validation grouped by state/district (GroupKFold [5]), to mimic a test set from unseen districts; every validation number below is out-of-fold (OOF) with the same decoding as the test run, the library in LOO mode and the null levels cross-fitted (two halves of the natural clips calibrate each other; synthetic clips use all natural clips). Fusion weights and

decode parameters were fitted on these same OOF predictions (no nested CV), so the OOF scores are optimistic for those hyperparameters.

II. Proposed Method

Background. Frame-level CRNN classifiers [6] are a standard approach to SED; Whisper's encoder keeps background-sound identity [7], and AudioSet [8] transformers (AST [9], BEATs [10], Dasheng [11]) are strong general audio encoders. The synthetic half is instead detection of a known signal, for which the matched filter is SNR-optimal in white Gaussian noise [13]; we use its normalised form, fast NCC [14].

Matched filter. For a test clip x and template T_i, the matcher computes the NCC $c_i(s)$ at every sample lag s with FFTs on the GPU. A per-template null level ν_i is the 0.999 quantile of $\max_s c_i(s)$ over the natural validation clips. Candidates are kept when the margin $c_i(s) - \nu_i \geq 0.1$ and selected greedily by margin with at most 30 % mutual overlap. In LOO mode on validation the matcher alone scores F1 0.9819 / Dice 0.9723 on the 695 synthetic clips (Table II); it fires on 687 of them and on 73 of the 1,806 natural clips.

SED ensemble. Each member is a BiGRU (hidden 192, dropout 0.3) with a linear layer to eight channels, trained with binary cross-entropy (masked to the clip's frames, per-channel weights as in the released code) and SpecAugment-style feature masking [15] on random crops, AdamW [16] (weight decay 10^{-2}) under a one-cycle schedule peaking at 10^{-3}; the weights after the last epoch are kept. Nine members keep the encoder frozen; each is the mean of its five fold heads: wsA (whisper-small, layers 3/6/9/12, validation only, 25 epochs, batch 16, 600-frame crops); wlP (whisper-large-v3, layers 8/16/24/32, the wsA recipe with pseudo-labels at weight 0.5); wlH, wlH2, wlH3 (large-v3, validation + Gold, 8 epochs, batch 8, 400-frame crops, seeds 0/1/2); wlHS (as wlH with Silver, 6 epochs); aH, bH, dH (AST / BEATs / Dasheng, layers 4/8/12, the wlH recipe). Three members update the encoder: aFG (all 12 AST blocks, encoder learning rate 2×10^{-5}, head 10^{-3}, 3 epochs, batch 8, fp16 autocast, gradient checkpointing, validation + Gold, 5 folds), bFG (the same on BEATs with the patch embedding frozen) and bFS (BEATs fine-tuned on all 2,501 validation clips + Gold + Silver, 2 epochs, batch 8, 6,320 steps, no hold-out). The ensemble probability is the weighted mean

$$p = \Big(\sum\nolimits_m w_m p_m\Big) \Big/ \sum\nolimits_m w_m, \tag{1}$$

with weights wsA 0.5, wlP 3, wlH 2, wlH2 6, wlH3 1, wlHS 6, aH 8, aFG 8, bH 8, bFG 8, dH 4, bFS 4. The weights of the eleven members with OOF predictions were set by coordinate descent over a coarse grid on the natural clips (`tune_weights.py`; the matcher dominates the synthetic ones); bFS's weight was a leaderboard probe. Members were added one at a time; our working rule was that a candidate had to raise the pooled OOF score with at least four of five district folds up before it was given one upload ("one-slot probe"), which decided; the per-member OOF records were not retained, so this is stated as practice, not demonstrated.

Decoding and fusion. Natural clips use two branches: $p_h = \min(p_{\text{hns}}, p_0)$ for human non-speech (hns) and $p_o = \min(\max_{c\neq\text{hns}} p_c, p_0)$ for the other categories, p_0 being the any-noise channel. Each branch is decoded peak-relatively: a region is detected where the median-filtered curve (window 15 / 7 frames) clears a floor of 0.2, is kept if its peak reaches 0.34 / 0.30, and both boundaries are placed where the curve crosses α times that event's own peak ($\alpha = 0.6$ / 0.5). This replaces one global threshold; when introduced (before upload 924220) it was worth +0.029 natural OOF event F1 on the same probabilities. Events shorter than 0.1 s are dropped, gaps below 0.1 s / 2.0 s are closed, and the branches are merged with duplicate removal at IoU 0.5. A clip with any library hit is treated as synthetic: its events are the template events plus any SED event (any-noise channel, threshold 0.7, median 3, minimum 0.1 s, gap 0.04 s) that does not overlap them (all parameters: `models/fusion_Q1_bFSadd4.json`).

Compute and software. Python 3.13.6, PyTorch 2.14.0 (CUDA 12.6). Frozen heads (on cached float16 features), all aFG folds and bFG folds 0–2 were trained on one RTX 4060 Laptop GPU (8 GB); bFG folds 3–4 and bFS on Kaggle T4 GPUs. Test inference: five frozen-backbone and eleven fine-tuned encoder passes (5 aFG, 5 bFG, bFS) over the 5,517 clips, and 84.6 min for the matched filter on the laptop GPU (FFT up to 2^{19}). Pretrained encoders: openai/whisper-small and whisper-large-v3 [17] (Apache 2.0), MIT/ast-finetuned-audioset-10-10-0.4593 [9] (BSD 3), BEATs iter3+ AS2M [10] (MIT), mispeech/dasheng-base [11] (Apache 2.0); all public. The original encoders are not redistributed; the eleven fine-tuned AST/BEATs checkpoints are released as derivatives under BSD 3 / MIT with the original notices.

III. Results and Conclusion

Leaderboard. Table I traces the system from the first upload (1.4591) to the final 1.6009, the highest of all our uploads listed on Codabench. The changes between the first upload and v14 (more members, Gold data, the split and peak-relative decodes) are not separable in our records and account for most of the gain (to 1.5808); fine-tuning AST, dropping aA and adding BEATs and Dasheng members add +0.0131; decoding changes tuned on OOF predictions add +0.0063; bFS adds +0.0007, below what we can distinguish from noise. The gap to rank 1 (1.6173) is 0.0164: event F1 0.0092, Dice 0.0072. Selection was leaderboard-guided throughout, so the final score carries an optimistic bias that we did not quantify. OOF and leaderboard scores of one configuration differ by 0.12 (1.4787 vs 1.6002); we attribute this to the test set being 39 % synthetic-like (2,164 of 5,517 clips have a library hit, against 28 % truly synthetic and 30 % with a LOO hit in validation) and to the test run using the full library.

Components. Table II separates the two branches: on the synthetic clips the matcher alone reaches 1.9542 and the SED alone 1.3378; on the natural clips the SED alone gives 1.1409, the matcher alone 0.0691, and the fusion adds +0.016 (template events in the 73 natural clips with a hit). wlP is worth +0.0028 overall. The natural clips remain the bottleneck

TABLE I

LEADERBOARD (LB) LINEAGE: 10 OF OUR AT LEAST 48 UPLOADS; THE OTHERS WERE NOT ADOPTED. OOF: OUT-OF-FOLD VALIDATION SCORE OF THE ROW'S CONFIGURATION (`RUN.PY TUNE`: LOO LIBRARY, CROSS-FITTED NULLS); ΔOOF: GAIN OVER THE PREVIOUS ROW (FOLDS UP OF 5), WHERE RECORDED. MED / THR: MEDIAN WINDOW (FRAMES) / DETECTION THRESHOLD OF THE HUMAN-NON-SPEECH BRANCH. Q1 = THE FINAL CONFIGURATION WITHOUT BFS.

Upload	Change	OOF	ΔOOF	LB F1	LB Dice	LB comb.
921278[c]	first upload (10 Sep)	–	–	0.7113	0.7478	1.4591
924220	v14: 6 Whisper + 2 AST frozen members	1.4466	–	0.7752	0.8055	1.5808
929944	+ aFG:8, aA dropped[b]	–	–	0.7739	0.8130	1.5868
932046	+ bH:8	–	–	0.7741	0.8148	1.5890
932698	+ bFG:8, dH:4	–	–	0.7752	0.8187	1.5938
933670	boundary calibration (med 11, thr 0.32)	–	+0.0024 (5)	0.7778	0.8194	1.5972
936587	med 13, thr 0.34	–	+0.0017 (4)	0.7799	0.8201	1.5999
991824	med 15	–	–	0.7802	0.8198	1.6000
991838	other-branch gap 2.0 (Q1)	1.4787	+0.0007 (4)	0.7806	0.8195	1.6002
994586	**+ bFS:4 (final)**	n/a[a]	–	**0.7803**	**0.8206**	**1.6009**

[a] bFS trains on all validation clips: no OOF estimate; its weight was a one-slot leaderboard probe. [b] aA, the second AST member of v14 (validation only), was dropped after its artifacts were lost; v14 without it scores 1.4474 OOF. [c] No longer listed on Codabench; values from our records. Differences below 0.001 between the last rows are smaller than we can distinguish from leaderboard noise, which we did not quantify.

TABLE II

OUT-OF-FOLD COMPONENT SCORES OF CONFIGURATION Q1 (THE FINAL SYSTEM WITHOUT BFS) ON VALIDATION: COMBINED (F1 / DICE) ON ALL 2,501 CLIPS, THE 1,806 NATURAL AND THE 695 SYNTHETIC CLIPS. LIBRARY IN LOO MODE, NULL LEVELS CROSS-FITTED.

System	All	Natural	Synthetic
Q1 full	1.4787 (.6894/.7894)	1.1570 (.4465/.7105)	1.9868 (.9925/.9943)
without wlP	1.4759 (.6872/.7887)	1.1527 (.4431/.7096)	1.9866 (.9923/.9943)
retrained aFG/bFG	1.4771 (.6886/.7885)	1.1542 (.4450/.7092)	1.9869 (.9925/.9944)
SED only	1.2005 (.4500/.7505)	1.1409 (.4331/.7077)	1.3378 (.4761/.8618)
Templates only	0.8871 (.5976/.2895)	0.0691 (.0424/.0268)	1.9542 (.9819/.9723)

(F1 0.45 against 0.99 synthetic): on a pre-v14 ensemble's natural OOF predictions only 12.9 % of reference events were truly missed while 52 % overlapped a prediction that failed a boundary tolerance, hence the focus on per-event boundary placement. Also tried and absent from the final system: a soft-Dice loss term (wlHSD, refuted out-of-fold), a stacker over the members, CLAP and CED [12] members, a fine-tuned Whisper (wFG), a fine-tuned Dasheng (dFG) and a second AST seed; their configuration files (`fusion_STK*`, `K2/K3`, `Q1_eH4`, `W*`, `G_*`, `D_*`) are in the repository; the basis of each rejection was not uniformly recorded.

Reproduction. After the final upload the workspace lost the backbone weights, the fine-tuned checkpoints, all cached test probabilities and the test template candidates; the 45 frozen fold heads, the configurations, the template library and the scored zip survived. We retrained aFG, bFG and bFS with the identical scripts and default seeds on Kaggle T4 GPUs (the originals of aFG and bFG folds 0–2 were trained on the laptop GPU), re-ran the matcher, and recomputed the frozen members from the released heads and re-downloaded public encoders. The retrained aFG/bFG give 1.4771 OOF against 1.4787 with the originals (Table II); the rebuilt submission flags the same 2,164 clips as synthetic-like, is identical on 41.8 % of the clips (2,008 of 2,164 synthetic-like, 298 of 3,353 natural-like; 13,442 vs 13,451 events) and scores 1.9858 (F1 0.9943 / Dice 0.9915) under the competition metric with the scored file as reference. The released fine-tuned checkpoints (eleven files, 3.96 GB, release assets linked from the repository) are thus not bit-identical to the scored ones (different GPU, nondeterministic kernels); bFS has no OOF estimate, so its equivalence rests on that agreement. Frozen members and matcher are deterministic up to floating-point differences of the encoder and FFT passes. The pseudo-label file and teacher heads of wlP are not released.

Limitations. The matched filter relies on the test set reusing the validation noise library; with fresh waveforms the system falls back to the SED branch (1.2005 OOF). The ensemble (five frozen backbones, eleven fine-tuned encoders) was chosen for score, not cost; wlP uses test audio transductively; natural-clip event F1 (0.45 OOF) is the weakest part.

Conclusion. Known-signal detection on the synthetic half and frame-level SED with per-event boundary placement on the natural half gave a 5th-place system, released with heads, configurations and both zips.

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